Quiz 2 — Sample 1

DATA 412/612

Write your answers in an R script. Use the tidyverse. Load `nycflights13` and the `flights` data. This practice quiz has questions only.

1. In `flights`, for every `origin`, find the number of `NA`s in each column that is numeric and starts with `arr`, or that is character and contains `ca` or `il`.

2. In `flights`, for every destination, find the number of flights that were delayed by more than 2 hours on arrival in the fall (September, October, or November) and that were operated by the least popular airline.

3. Between which two airports did the worst delayed flights occur? Treat a flight as delayed when `arr_delay > 0` and `dep_delay > 0`. How many delayed flights occurred between those two airports?

4. Which three carriers have the highest rates of departure delays exceeding 15 minutes? What are those proportions, compared with the total number of flights each carrier operated?

5. Use `readr` parse functions to correct the type of each column in the data frame below. The column names are a hint. Treat the strings `N--a` and `N-a` as missing.

```
df <- data.frame(
  date = c("2023--11-21", "2023--09-02", "2020--01-30"),
  time = c("01::00-00", "13::58-01", "23::00-09"),
  num  = c("10.5", "20.3", "15.8"),
  log  = c("TRUE", "N--a", "FALSE"),
  char = as.factor(c("Apple", "Banana", "N-a")),
  fac  = c("A", "B", "C")
)
```